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## Pathogens associated with diarrhea in the GEMS1 study

### *An exploratory data analysis exercise*

Goal: In this exercise you will perform a step-by-step exploratory data analysis on the ClinEpiDB platform to explore pathogens associated with diarrhea in the GEMS1 Case Control study.

#### Step 1: Read the study page and formulate a hypothesis

Go to the [GEMS1 Case Control study page](#). Click on the View study details tab and read the summary and description of this observational study conducted in 7 sites in Asia and Africa. This is a case control study of children under 5 years of age where cases had moderate-to-severe diarrhea and controls were diarrhea-free. Stool samples from cases and controls were compared to identify the etiology (causes) of diarrhea.

**Hypothesis:** In infants under 1 year of age in India, rotavirus, Cryptosporidium and Giardia infection are associated with moderate-to-severe diarrhea.

Give your analysis a name at the top of the page. It may look something like this.

### GEMS1 Case Control

Causes of diarrhea in infants, India



#### Step 2: Choose an appropriate subset of data

Click the Browse and subset tab. If you want to restrict your analysis to participants under 1 years of age, and to participants in India, how would you choose an appropriate subset of data?

Country

View Study Details [Browse and Subset](#) Visualize Download Record Notes

▼ Featured variables

Household

- Country

Household repeated measure

- Household study timepoint

Participant

- Age group at enrollment
- Case or control participant

Participant repeated measure

- Age
- Study timepoint

expand all | collapse all

Find a variable

Household

- Geographic location
  - Country

### Country

Original variable name: SITE

Keep checked values at top

1 row selected.

|                                     | Country    | Subset of Households |       |
|-------------------------------------|------------|----------------------|-------|
| <input type="checkbox"/>            | Bangladesh | 3,859                | (17%) |
| <input checked="" type="checkbox"/> | India      | 3,582                | (16%) |
| <input type="checkbox"/>            | Kenya      | 3,359                | (15%) |
| <input type="checkbox"/>            | Mali       | 4,097                | (18%) |
| <input type="checkbox"/>            | Mozambique | 1,976                | (9%)  |
| <input type="checkbox"/>            | Pakistan   | 3,096                | (14%) |
| <input type="checkbox"/>            | The Gambia | 2,598                | (12%) |

## GEMS1 Case Control

Causes of diarrhea in infants, India

Country Age group at enrollment

View Study Details [Browse and Subset](#) Visualize Download Record Notes

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Household

- Country

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Participant

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Participant repeated measure

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expand all | collapse all

### Age group at enrollment

Original variable name: agegroup

Keep checked values at top

1 row selected.

|                                     | Age group at enrollment | Subset of Participants |        |        |
|-------------------------------------|-------------------------|------------------------|--------|--------|
| <input type="checkbox"/>            |                         | 3,582                  | (100%) | 22,567 |
| <input checked="" type="checkbox"/> | 0-11 months             | 1,357                  | (38%)  | 8,906  |
| <input type="checkbox"/>            | 12-23 months            | 1,186                  | (33%)  | 7,586  |
| <input type="checkbox"/>            | 24-59 months            | 1,039                  | (29%)  | 6,075  |

How many participants are present in your subset? \_\_\_\_\_

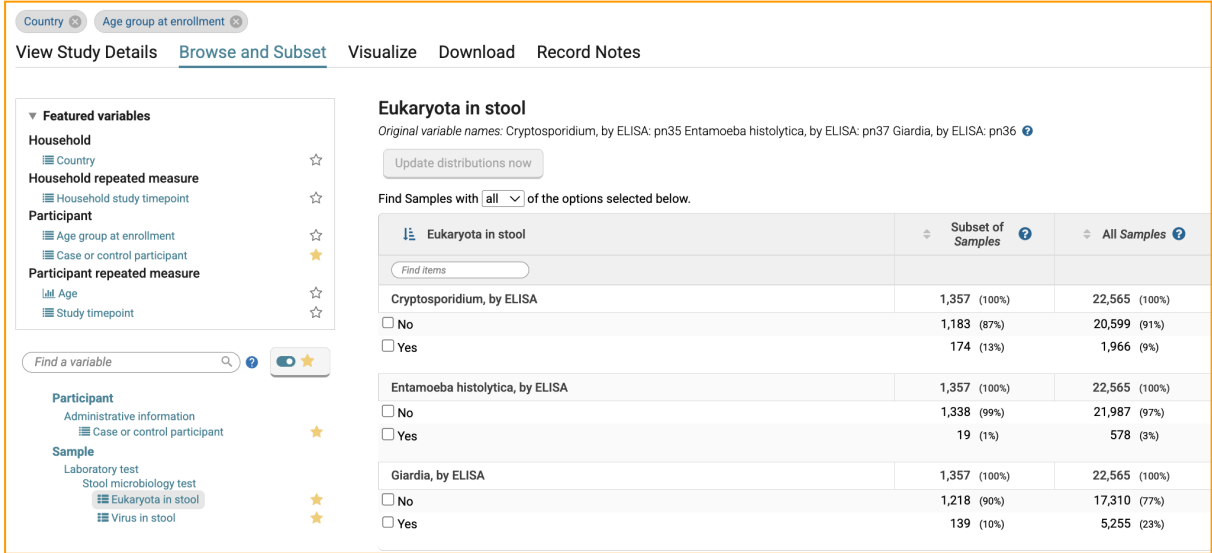
Looking at the dataset diagram at the top of the page will indicate that the subset includes 1,357 participants out of the 22,567 participants in the study.

### Step 3: Identify variables of interest for this analysis

Browse or search through the variable tree on the left and identify variables that will be interest in your analysis. Look at the distribution of each variable and note whether it is numeric or categorical, as this will help you decide what visualization tools to use in the next step. Star the variables of interest to make them easier to access.

**Case or Control status:** We want to compare pathogen prevalence between cases and controls, so we will need the variable *Case or control participant*, a categorical variable.

**Pathogens:** Pathogens are detected by tests performed on samples (stool sample in the case of diarrhea), so look under the Sample category in the variable tree on the left. There are a number of pathogen test results under Sample > Laboratory test > Stool microbiology test. Two of the pathogens we are interested in, *Cryptosporidium* and *Giardia*, are Eukaryotes and the third, rotavirus, is a virus. The variables needed to test our hypothesis are *Cryptosporidium, by ELISA* and *Giardia, by ELISA* under Eukaryota in stool and *Rotavirus, by ELISA* under Virus in stool . They are all categorical variables.



**Step 4: Create visualizations to examine associations between variables**  
Make a list of the associations you would like to plot. What variables do you want to plot on the X-axis and on the Y-axis? What sort of plot would be appropriate for these variables?

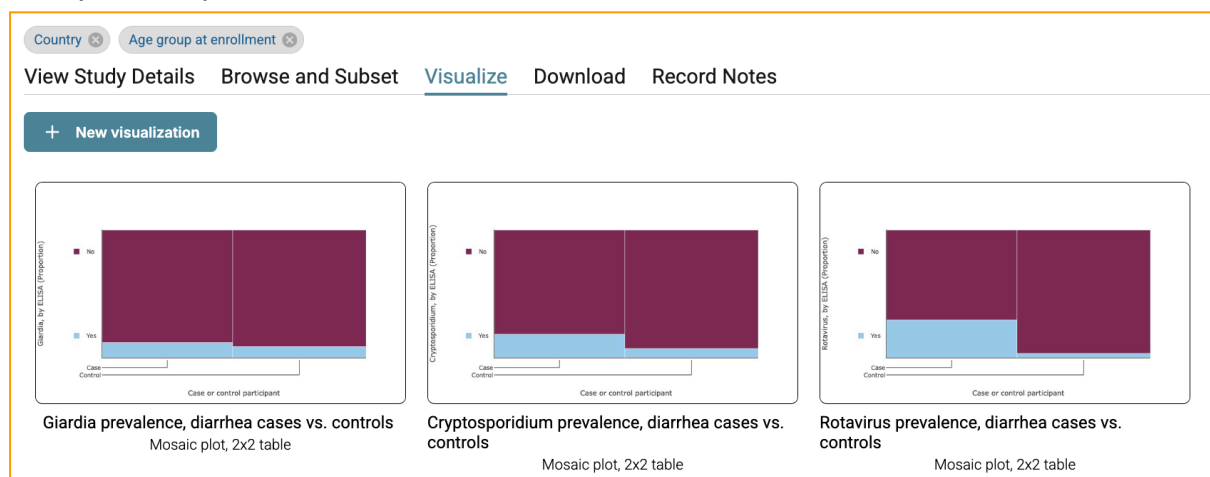
| Association                           | X axis | Y axis | Plot |
|---------------------------------------|--------|--------|------|
| Rotavirus in cases and controls       |        |        |      |
| Cryptosporidium in cases and controls |        |        |      |
| Giardia in cases and controls         |        |        |      |

Your plan may look like this. Both the X-axis and Y-axis variables for each association are binary categorical variables, so a 2x2 table would be appropriate to explore this association.

| Association                           | X axis                                                                     | Y axis                                                                | Plot                     |
|---------------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|--------------------------|
| Rotavirus in cases and controls       | <i>Case or control participant</i><br>(categorical variable with 2 levels) | <i>Rotavirus, by ELISA</i><br>(categorical variable with 2 levels)    | Mosaic Plot<br>2x2 Table |
| Cryptosporidium in cases and controls | <i>Case or control participant</i><br>(categorical variable with 2 levels) | <i>Cryptosporidium, by ELISA</i> (categorical variable with 2 levels) | Mosaic Plot<br>2x2 Table |
| Giardia in cases and controls         | <i>Case or control participant</i><br>(categorical variable with 2 levels) | <i>Giardia, by ELISA</i> (categorical variable with 2 levels)         | Mosaic Plot<br>2x2 Table |

Click on the Visualize tab , then on new visualization, and select the appropriate tool and make the plots. Name each plot.

Your plots may look like this:



Interpret the plots. What does the data say about your hypothesis?

The 2x2 mosaic plots indicate the following about participants in India under 1 year of age-

- a) Rotavirus: 29.6% of diarrhea cases have rotavirus as compared to 3.2% of controls, so rotavirus infection is associated with moderate-to-severe diarrhea, supporting our hypothesis.
- b) Cryptosporidium: 18.5% of diarrhea cases have rotavirus as compared to 7.3% of controls, so Cryptosporidium infection is associated with moderate-to-severe diarrhea, supporting our hypothesis.
- c) Giardia: 11.9% of diarrhea cases have rotavirus as compared to 8.6% of controls, so Giardia infection is not as strongly associated with moderate-to-severe diarrhea.

This interpretation of our exploratory data analysis is supported by the published results of the GEMS1 study showing that rotavirus and Cryptosporidium are associated with moderate-to-severe diarrhea in children while Giardia is associated with asymptomatic colonization.

*Thank you for completing this exercise on performing an exploratory data analysis on [clinepidb.org](http://clinepidb.org)!*

*Please contact [help@clinepidb.org](mailto:help@clinepidb.org) with feedback or questions.*

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